

Does Anything Survive an Honest Backtest?

Falsification-First Evidence on Cross-Sectional Anomalies in Equities, Crypto, and Closed-End Funds

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ABSTRACT

I ask whether the standard published price signals still pay, in the markets a free-data researcher can reach, once every removable form of backtest inflation is removed. Across eleven pre-registered trials spanning three asset classes the answer is largely no, and the manner of the failure is the contribution. On a survivorship-biased universe the pipeline yields a net Sharpe of 0.82; on a point-in-time universe the identical code yields -0.01 — the apparent alpha was the bias. Seven equity trials, spanning horizons, labels, neutralization, and three model classes, never clear a Deflated Sharpe of 0.04. Extending the same machinery to crypto-perpetual funding carry recovers a real, strongly significant premium ($t = -3.5$, net Sharpe 0.87) that nonetheless fails the pre-registered evidence bar (Deflated Sharpe 0.865), is severely left-skewed, and has decayed from a Sharpe of 2.3 to roughly 0.4 as the trade institutionalized; pushed into the liquid tail the same funding signal is as significant ($t = -3.6$) yet loses money net, the cleanest demonstration that significance and tradability are distinct. A matched-control study shows the post-effective S&P 500 deletion rebound is small-loser mean reversion, not an index effect. A closed-end-fund discount-reversion strategy then cleared every pre-registered bar at once (net Sharpe 1.1, Deflated Sharpe 0.999) before a one-week entry lag collapsed it to nothing — the discipline overturning its own best result. The deliverable is a research process that detects a genuine premium where one exists, declines to claim it where one does not, and overturns its own result when it is too good to be true.

Keywords: market efficiency; cross-sectional anomalies; survivorship bias; deflated Sharpe ratio; funding-rate carry; research integrity. **JEL:** C58, G11, G12, G14.

1. Introduction

Can the most widely published price-only cross-sectional signals — 12-1 and 6-1 momentum, short-term reversal, low volatility, distance from the 52-week high — be traded profitably in the S&P 500 today, net of costs, after one removes every backtest inflation available to a free-data researcher? The published record predicts not, and I confirm it. The harder question is whether a null can be demonstrated convincingly enough to survive adversarial scrutiny, and whether the same machinery, confronted with a market in which a premium genuinely exists, both finds it and is refused by its own pre-registered standard.

I make three contributions. First, I reproduce McLean and Pontiff (2016) in-house: correcting a single source of bias — the survivorship of the universe — collapses a net Sharpe of 0.82 to -0.01 under otherwise identical code. Second, I show across two asset classes that the discipline holds when the number is good: crypto funding carry produces a real signal that I decline to claim because it fails the Deflated Sharpe bar and is structurally crash-prone. Third, the entire apparatus is falsification-first — a planted-signal / pure-noise harness gates every result in continuous integration — and I document the artifacts it caught during development, several of which would have produced false positives.

2. Why the evidentiary bar is set where it is

Four findings dictate the design, and I treat them as constraints rather than suggestions. (i) Published anomaly returns fall 26% out-of-sample and 58% post-publication (McLean and Pontiff, 2016). (ii) At honest multiple-testing thresholds, most

published factors are likely false discoveries (Harvey, Liu, and Zhu, 2016); I therefore count every trial and never reset the count, which presently stands at $N = 11$ and feeds the Deflated Sharpe Ratio directly. (iii) Anomaly profits concentrate in high-turnover implementations whose costs exceed their gross returns (Novy-Marx and Velikov, 2016); no gross-only number is reported and turnover is a headline statistic. (iv) The expected maximum Sharpe of N noise strategies grows with N (Bailey and Lopez de Prado, 2014); the Deflated Sharpe benchmarks each result against that expected maximum at the true N .

3. Data and the survivorship exhibit

Daily adjusted prices are obtained via yfinance over 2009–2026. Point-in-time S&P 500 membership is reconstructed by walking the index changes table backward from current constituents: 810 names held a seat at some point in 2010–2026, of which 661 (81.6%) have retrievable price history. Features are masked to members-as-of-date before cross-sectional normalization, so a name absent from the index cannot shift the statistics the model observes.

The static present-day universe is retained as a measured exhibit. Under identical code, features, and costs, the only change being the universe definition, the result inverts:

Table 1. The same pipeline on a biased versus an honest universe.

	Static (biased)	Point-in-time (honest)
Mean rank IC	0.0333	0.0052
Net Sharpe	0.82	-0.01
Equal-weight long-only Sharpe	1.00	0.81

Survivorship bias did not inflate the result; it was the result. A universe of known survivors renders “buy the dip” unfalsifiable, since every drawdown in the panel is one the firm lived through. I bound the residual bias rather than impute it: forcing a synthetic -30% final return onto the 29 names that die mid-window moves the net Sharpe by +0.006 against a zero-return control, so the missing delisting returns can neither rescue nor sink the result. Imputation is avoided on principle — delistings are missing because the firm failed, so any model fit on survivors re-injects the very bias just removed. The 149 never-priced names are the unbounded residual and require CRSP-grade data.

4. Methodology

Validation. Expanding walk-forward with an embargo of at least the label horizon: a model trained through date t observes no row whose 21-day forward label overlaps the test window. k -fold cross-validation is inadmissible on financial panels, since random folds place future observations in the training set of past ones.

Falsification gate. Before any real-data result is read, the pipeline must recover a synthetic planted signal (Deflated Sharpe 0.992) and reject a pure-noise panel of identical shape (0.078). Both run in continuous integration on every commit; a change that introduces leakage causes noise mode to “find alpha” and fails the build.

Inference. Daily rank ICs of overlapping 21-day labels share 20 days of information; treating them as independent overstates significance by roughly the square root of the horizon. On the planted panel a naive t of 7.76 falls to a Newey-West t of 2.00, as theory predicts. All reported t -statistics are Newey-West with lags equal to the horizon.

Construction, costs, and baselines. Decile dollar-neutral long-short, 10 bps linear cost on turnover, sector-demeaned and projected to zero ex-ante beta with rolling past-only estimates; realized beta drift is measured (mean 0.05, 95th percentile 0.23) rather than assumed zero. Every model must beat equal-weight and a one-line 12-1 momentum rank on identical dates and costs. On the planted panel the one-line baseline beats the five-feature ridge, 1.19 to 0.86 — a multi-feature model is not automatically superior to its strongest ingredient.

5. Results

5.1 Equity price features

No equity configuration produces a defensible edge. The best Deflated Sharpe is 0.04 and no correctly-signed Newey-West t reaches 2. Because linear, tree, and shallow-network learners on identical features and harness are uniformly null — and each recovered the planted signal beforehand — the null is localized in the features, not the learner.

Table 2. Seven equity trials, point-in-time universe, net of costs, at the true N for the Deflated Sharpe (DSR).

#	Specification	IC	t_{NW}	Net SR	DSR	Turn.
1	Static universe (biased exhibit)	0.033	—	+0.82	0.998	3.8x
2	Point-in-time universe	0.005	+0.54	-0.01	0.29	7.3x
3	+ sector / beta neutralization	0.005	—	-0.38	0.01	7.4x
4	Quarterly horizon (turnover attack)	-0.028	-1.95	-0.35	0.01	2.4x
5	Residualized label	+0.023	+1.91	-0.77	~0	3.5x
6	Gradient boosting	+0.008	+0.80	-0.12	0.04	5.4x
7	Shallow neural network	+0.009	+1.21	-0.28	0.01	5.8x

5.2 Cross-asset extension: funding-rate carry

To address the concern that a uniformly null pipeline may simply be unable to find anything, I extend the identical machinery to perpetual-futures funding carry — a premium with a structural origin (payment to the party warehousing crowded leveraged-long demand), on a universe with no survivorship gap, since delisted contracts retain full terminal histories. The label is funding-inclusive total return; a price-only label is shown on synthetic data to reject a true premium.

Table 3. Trial 8 — crypto funding carry against pre-registered criteria (top-30 by dollar volume, weekly, 7 bps per side, $N=8$).

Criterion	Value	Bar	Pass
IC Newey-West t	-3.54	≤ -2	Yes
Net Sharpe	+0.87	> 0	Yes
Deflated Sharpe ($N=8$)	0.865	≥ 0.95	No
Survives excluding top 3 names	0.68	> 0	Yes
Shuffled-funding control	0.08	~ 0	Yes
Skew / max drawdown	-1.87 / -74%	—	(crash-prone)

The single failed criterion is the Deflated Sharpe, and it was not relaxed; “0.865 is close” is precisely the rationalization the protocol forbids. Diagnostics confirm the signal is genuine rather than an artifact: delaying entry one to five days decays the Sharpe gracefully (0.87 to 0.39) rather than collapsing, which excludes a timing leak; only two of twenty-eight delisted contracts were held at expiry, excluding delisting optimism; and the shuffled-funding control is flat. The premium is real but decaying — a Sharpe of 2.28 in 2020–21 falling to roughly 0.4 thereafter as basis-trade capital scaled — and its -1.87 skew is the structural cost of selling funding. This is the strongest evidence in the study: the pipeline detects a real premium where one exists, implying the seven equity nulls reflect genuine absence rather than incapacity.

5.3 The disappearing deletion effect

A matched-control event study on 75 discretionary S&P 500 deletions (2010–present) asks whether deleted names out-rebound a basket matched on size and trailing return over the 60 days after the effective date. They do not: the daily event-time portfolio has a net Sharpe of -0.04, a Newey-West t of -0.10, and a Deflated Sharpe of 0.05. Deleted names do rebound (about +4.8% over 60 days), but a matched control rebounds +2.6% of that, leaving a +2.2% residual that is insignificant ($t = 0.87$) and negative before 2015. The rebound is small-loser mean reversion, not an index-deletion effect — Greenwood and Sammon's (2025) disappearing index effect reproduced in the post-effective window, with the control basket as the discipline that separates a real anomaly from a mechanical one.

5.4 The liquid-tail carry: significance without tradability

Carry was real in the top-30; trial 10 asks whether a wider premium hides in the liquid tail (dollar-volume ranks 31–150), beneath the basis-trade capital that decayed the majors. Pre-registered on the disjoint tail universe at conservative 20 bps-per-side fills, the funding signal predicts the cross-section as strongly as the majors — a Newey-West t of -3.62 versus -3.54 — yet loses money net: a Sharpe of -0.13 on a gross of 0.26 . The decomposition is the lesson: funding income contributes $+1.23$ in cumulative P&L while price drift returns -0.85 of it, so the tail carry is almost fully priced and the fills finish it. It is the cleanest illustration in the study that a strongly significant signal and a tradable edge are different objects.

5.5 A graduation overturned: closed-end-fund discount reversion

The one structurally protected premium remaining was the closed-end-fund discount: with no creation or redemption mechanism, price can persist away from NAV, and in the sub-\$400M tail no arbitrageur is large enough to close it. A zero-trial Stage 1 first cleared the survivorship question the opposite way — 151 dead closed-end funds enumerated from SEC filings, 94% dying at NAV (liquidation, merger, open-ending, term maturity) with no distress delistings — so omitting dead funds biases a discount-long book against itself. The registered Stage 2 then passed every pre-registered criterion at once: net Sharpe 1.11 , Deflated Sharpe 0.999 at $N=11$, a Newey-West t of -10.4 , positive skew. The prime directive treats a result that good as a suspected defect, so an entry-lag sweep was run: the Sharpe collapsed from 1.11 to 0.10 at one week of delay and turned negative beyond, exposing a one-week microstructure bounce — the discount in week w is built from the same close the next week's return divides by — rather than durable reversion. The implementable strategy is null and the criteria were not relaxed; the lesson is now a registered requirement, that a reversion strategy must pass an entry-lag implementability gate and not merely the in-sample bar.

6. What the falsification harness caught

Several development-stage artifacts would have produced false positives. A baseline expected to be near 0.9 reading an equal-weight Sharpe of 3.34 exposed a return-construction bug in which pad-filled delisted names contributed frozen zero returns and crushed measured volatility. On signal-free synthetic data, residualized-label momentum carried a volatility-regime-dependent IC of $+0.06$ to $+0.13$ from estimation error alone — an artifact that masquerades as “momentum works in calm regimes,” a hypothesis I had registered, and which now requires a paired control before any run. In three separate settings absolute statistics misled where paired controls did not, which is now standing practice. A carry-book weighting bug that accumulated stale positions was caught in pre-trial review rather than after the fact. Finally, diffing successive vendor downloads of the same history reveals that “point-in-time” has a data-values dimension: the vendor re-serves history with float-level noise and occasional large corrections, so a backtest and a live model train on materially different versions of the past.

7. Capacity and execution

A square-root market-impact sweep on the deployed equity configuration (spread 10 bps, ADV coverage 87.6%) gives annual cost drag of 1.36% at $\$1M$, 2.27% at $\$10M$, 5.16% at $\$100M$, and 14.30% at $\$1B$. Because the gross edge is negative the formal capacity is zero; the informative object is the drag curve, which states that any strategy at this turnover ($3.46\times$ per year) needs at least 1.4% of true gross annual alpha to exist at $\$1M$ and 5% at $\$100M$. The carry strategy, at $35\times$ annual turnover, is a second reason it does not graduate beyond a logged result.

8. Live verification

The best honest equity configuration is paper-traded against a broker API — not in expectation of profit, which the evidence excludes, but because live information coefficient versus backtest information coefficient is the ultimate out-of-sample test of the pipeline. A daily job rebuilds the universe, trains only on fully realized labels, writes the full prediction cross-section to a write-once log before any order exists, and submits capped, integer-share orders to a paper endpoint. A control arm shadow-logs the momentum baseline on the same names, so a future divergence can be attributed to the model rather than the period. The first interpretable comparison matures roughly 21 trading days after the first cycle;

early P&L is explicitly not interpreted. Two unbackfillable datasets — daily data-revision fingerprints and a short-borrow-fee cross-section — accrue in parallel.

9. Limitations and the path to institutional grade

Residual survivorship bias remains (149 unpriceable names; missing delisting returns, bounded at 0.006 Sharpe for priceable failures); sectors are as-of-today; betas are estimated; headline costs are linear with impact priced separately; and the evidence spans three asset classes — equities and crypto over roughly fifteen and six years, and closed-end funds over fourteen years of weekly data. An institutional-grade version would require CRSP-quality delisting returns, point-in-time fundamentals and classifications, borrow availability and fees on the short book, an impact model calibrated to fills, multi-market replication, and an order of magnitude more trials under the same logging discipline. That the binding constraint is now data is itself a measured finding: a free historical ticker-to-CIK recovery over the dead names lifts survivorship-safe fundamentals coverage only from 73% to 75%, most resolved names carrying ticker-reassignment risk, so the quality-fundamentals hypothesis waits on institutional data rather than another free-data trial. Each limitation is named deliberately; naming them is part of the result.

10. Conclusion

I tested eleven pre-registered hypotheses across three asset classes and graduated none to production. Read carelessly that is failure; read correctly it is the point. A research process is valuable only if it can distinguish a real premium from a lucky one and a robust premium from a decaying, skewed one, and then act on the distinction even when the number in front of it is good. The pipeline destroyed its own best result upon correcting the universe, recovered genuine crypto carry yet declined it on a deflated-Sharpe standard it was right to enforce, showed a famous anomaly to be a matched-control artifact, and overturned its own best-looking backtest — a closed-end-fund reversion that cleared every bar — on a one-week implementability test. The execution and risk engine that would size any graduated edge is built and shown on synthetic ground truth to commit to a real signal and size a null to zero; the missing piece is the edge. The strategies failed; the judgment did not.

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* Independent undergraduate research project. All code, data, the complete trial log, and the live experiment are public. ASU theming is visual only and implies no institutional endorsement; this is a working paper and has not been peer reviewed. Every figure traces to a logged trial or a reproducible artifact in the project repository.